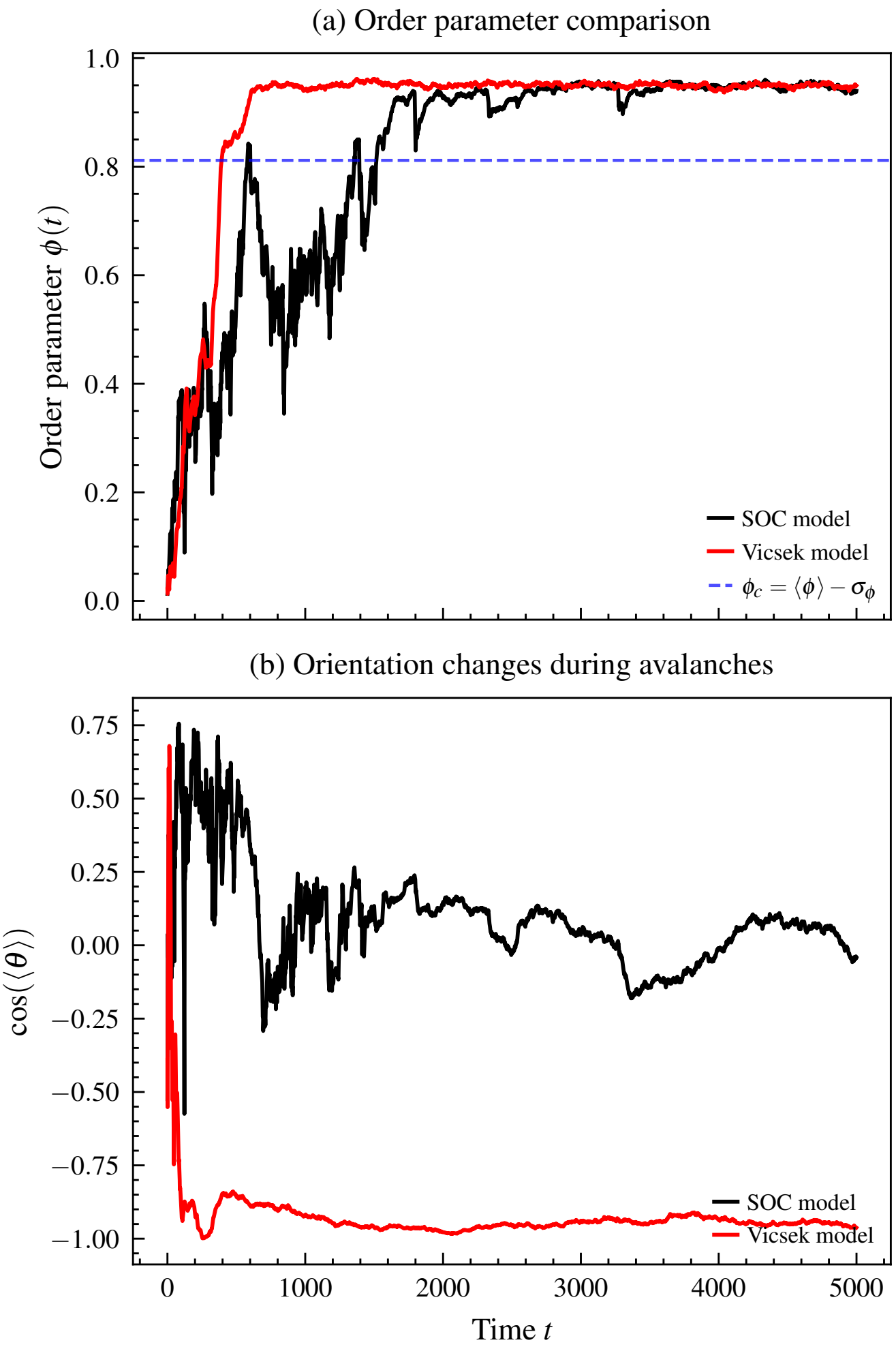
(a) Order parameter comparison
Order parameter $\phi(t)$
1.0
0.8
0.6
0.4
0.2
0.0
SOC model
Vicsek model
$\phi_c = \langle\phi\rangle - \sigma_\phi$
(b) Orientation changes during avalanches
$\cos(\langle\theta\rangle)$
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
SOC model
Vicsek model
0
1000
2000
3000
4000
5000
Time t